



UNSW
SYDNEY

MATH1231 Mathematics 1B – Algebra

Section 7: Linear transformations

Lecturer: Aidan Sims (aidan.sims@unsw.edu.au)

Slides: David Angell, Laure Helme-Guizon and others

Investigation

Exercise 1. Looking for similarities.



Expand the following expressions:

a) $\frac{d}{dx}(p + q) =$

$$\frac{d}{dx}(\lambda p) =$$

b) A is a matrix. $A(\vec{x} + \vec{y}) =$

$$A(\lambda \vec{x}) =$$

c) $\int_0^1 f(t) + g(t) dt =$

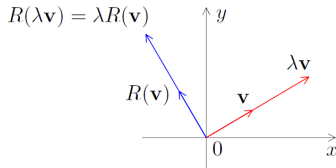
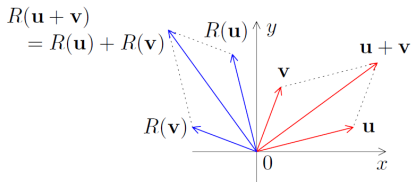
$$\int_0^1 \lambda f(t) dt =$$



Question. What do you notice about these problems?

Example 2: Rotation.

If $\vec{v}$ is any vector in $\mathbb{R}^2$, let $R(\vec{v})$ be the vector obtained when $\vec{v}$ is rotated by a right angle about the origin.



The above diagrams should (intuitively) convince you that

$$R(\vec{u} + \vec{v}) = R(\vec{u}) + R(\vec{v}) \quad \text{and} \quad R(\lambda \vec{v}) = \lambda R(\vec{v})$$

for all $\vec{u}, \vec{v}$ in $\mathbb{R}^2$ and all scalars λ .

Exercise 3. Use geometric arguments to carefully prove these two statements.

What did we learn from these examples?



As we saw earlier in connection with vector spaces, if essentially identical situations occur in many different guises, then it is worth our while to look at a more general concept which includes all these situations. This saves us work since we can solve a problem once instead of doing it many times, and it also gives us a deeper understanding of these problems by focusing our attention on what is really important about them.

Definition (linear transformation)

Let V and W be vector spaces over the same field $\mathbb{F}$. A function $T : V \rightarrow W$ is called a **linear transformation** if the following conditions are satisfied:

- T *preserves addition*: for all $\vec{u}, \vec{v}$ in V we have
$$T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v});$$
- T *preserves scalar multiplication*: for all $\vec{v}$ in V and all scalars λ we have

$$T(\lambda \vec{v}) = \lambda T(\vec{v}).$$

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Terminology.

- A linear transformation may be referred to as a *linear map* or a *linear mapping*; or just say “ T is *linear*”.
- Instead of “ T preserves addition” we sometimes say “ T satisfies the *addition condition*”.

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Terminology.

- A linear transformation may be referred to as a *linear map* or a *linear mapping*; or just say “ T is *linear*”.
- Instead of “ T preserves addition” we sometimes say “ T satisfies the *addition condition*”.

But **don't** say “ T is closed under addition” – the word “closed” means something quite different, as you will see if you carefully compare the definitions.



Now that we know what a linear map is, let's revisit our previous examples.

Example 1. Differentiation. Consider

$$T : \mathbb{P} \rightarrow \mathbb{P} \quad \text{where} \quad T(p) = \frac{d}{dx}(p).$$

Then T is a linear transformation; the two conditions in the definition express well-known facts about differentiation.

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Then T is a linear transformation; the two conditions in the definition express well-known facts about differentiation.

Example 2. Multiplication by a matrix. Let A be an $m \times n$ matrix and define

$$T : \mathbb{R}^n \rightarrow \mathbb{R}^m \quad \text{where} \quad T(\vec{x}) = A\vec{x}.$$

From basic properties of matrix algebra we have

$$T(\mathbf{x} + \mathbf{y}) = A(\mathbf{x} + \mathbf{y}) = A\mathbf{x} + A\mathbf{y} = T(\mathbf{x}) + T(\mathbf{y})$$

$$T(\lambda\mathbf{x}) = A(\lambda\mathbf{x}) = \lambda(A\mathbf{x}) = \lambda T(\mathbf{x})$$

for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and all $\lambda \in \mathbb{R}$; therefore T is a linear transformation.

Note that in this case the domain $V = \mathbb{R}^n$ and the co-domain $W = \mathbb{R}^m$ are different vector spaces.

Let's revisit our previous examples, continued.

Example 3. Taking the integral. Let $V = \{ \text{continuous functions on } [0, 1] \}$ and define

$$T : V \rightarrow \mathbb{R} \quad \text{where} \quad T(f) = \int_0^1 f(t) dt .$$

Once again, familiar facts about integration prove that T is a linear mapping.

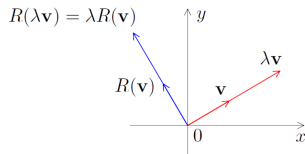
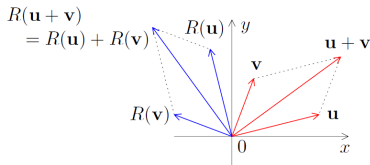
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Once again, familiar facts about integration prove that T is a linear mapping.

Example 4. Rotation. The rotation mapping from page 3 has $V = W = \mathbb{R}^2$, and $R(\vec{v})$ is the vector obtained by rotating $\vec{v}$ anticlockwise through a right angle. As we saw above, $R(\vec{u} + \vec{v}) = R(\vec{u}) + R(\vec{v})$ and $R(\lambda \vec{v}) = \lambda R(\vec{v})$ for all $\vec{u}, \vec{v} \in \mathbb{R}^2$ and all real λ ; this shows that R is linear.



Have I seen you somewhere?

Example 5. Let T be a function from $\mathbb{R}^2$ to $\mathbb{R}^3$ defined by

$$T \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} x_1 + 3x_2 \\ 5x_1 - x_2 \\ x_2 \end{pmatrix}.$$

Prove that T is a linear transformation.

Worked solution.

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Prove that T is a linear transformation.

Worked solution. • First we note that the domain and codomain of T are both vector spaces.

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Example 9. Let T be a function from $\mathbb{R}^2$ to $\mathbb{R}^3$ defined by

$$T \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} x_1 + 3x_2 \\ 5x_1 - x_2 \\ x_2 \end{pmatrix}.$$

Prove that T is a linear transformation.

Worked solution. • First we note that the domain and codomain of T are both vector spaces. • Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^2$. Then

$$\begin{aligned} T(\mathbf{x} + \mathbf{y}) &= T \begin{pmatrix} x_1 + y_1 \\ x_2 + y_2 \end{pmatrix} = \begin{pmatrix} (x_1 + y_1) + 3(x_2 + y_2) \\ 5(x_1 + y_1) - (x_2 + y_2) \\ x_2 + y_2 \end{pmatrix} \\ &= \begin{pmatrix} x_1 + 3x_2 + y_1 + 3y_2 \\ 5x_1 - x_2 + 5y_1 - y_2 \\ x_2 + y_2 \end{pmatrix} = \begin{pmatrix} x_1 + 3x_2 \\ 5x_1 - x_2 \\ x_2 \end{pmatrix} + \begin{pmatrix} y_1 + 3y_2 \\ 5y_1 - y_2 \\ y_2 \end{pmatrix} = T(\mathbf{x}) + T(\mathbf{y}) \end{aligned}$$

and so T preserves addition.

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Example 11. Let T be a function from $\mathbb{R}^2$ to $\mathbb{R}^3$ defined by

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Prove that T is a linear transformation.

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and so T preserves addition. • For any $\mathbf{x}$ in $\mathbb{R}^2$ and any scalar λ we have

$$T(\lambda \mathbf{x}) = T \begin{pmatrix} \lambda x_1 \\ \lambda x_2 \end{pmatrix} = \begin{pmatrix} (\lambda x_1) + 3(\lambda x_2) \\ 5(\lambda x_1) - (\lambda x_2) \\ \lambda x_2 \end{pmatrix} = \begin{pmatrix} \lambda(x_1 + 3x_2) \\ \lambda(5x_1 - x_2) \\ \lambda x_2 \end{pmatrix} = \lambda \begin{pmatrix} x_1 + 3x_2 \\ 5x_1 - x_2 \\ x_2 \end{pmatrix} = \lambda T(\mathbf{x})$$

and so T preserves scalar multiplication.

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Example 13. Let T be a function from $\mathbb{R}^2$ to $\mathbb{R}^3$ defined by

$$T \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} x_1 + 3x_2 \\ 5x_1 - x_2 \\ x_2 \end{pmatrix}.$$

Prove that T is a linear transformation.

Worked solution. • First we note that the domain and codomain of T are both vector spaces. • Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^2$. Then

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and so T preserves scalar multiplication. Therefore T is a linear transformation.

Have I seen you somewhere? YES, I have

Reflections on example 13. Let T be a function from $\mathbb{R}^2$ to $\mathbb{R}^3$ defined by

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Prove that T is a linear transformation.

Some comments on this example.

- For this kind of problem, do one step at a time, and make sure you don't leave out any steps.
- Be careful to do operations in the right order. This is easy if you look closely at the brackets – for example, to find $T(\vec{x} + \vec{y})$ you must *first* add $\vec{x}$ and $\vec{y}$, and *then* calculate T of the sum.

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- This example is really just our previous example with $A = \begin{pmatrix} 1 & 3 \\ 5 & -1 \\ 0 & 1 \end{pmatrix}$

Let A be an $m \times n$ matrix. Then

$$T : \mathbb{R}^n \rightarrow \mathbb{R}^m \quad \text{where} \quad T(\vec{x}) = A\vec{x}$$

is a linear map.



Example 14. Consider the function

$$F : M_{n,n} \rightarrow M_{n,n} \quad \text{where} \quad F(A) = A - A^T,$$

where A^T is the transpose of A (We call this function F to avoid any confusion between T for transformation and T for transpose.) Is this function a linear map?

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Then, for any A, B in $M_{n,n}$, we use properties of transposes (MATH1131, Chapter 5) to obtain

$$\begin{aligned} F(A + B) &= (A + B) - (A + B)^T \\ &= (A + B) - (A^T + B^T) \\ &= (A - A^T) + (B - B^T) \\ &= F(A) + F(B) \end{aligned}$$

so F preserves addition.

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so F preserves addition. Similarly, for any $A \in M_{n,n}$ and any scalar λ we have

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so F preserves scalar multiplication. Hence F is linear.

Example 7. Let $T : \mathbb{R} \rightarrow \mathbb{R}$ where $T(x) = 2x + 5$. Is this function a linear map?

$$T(1) = 7 \quad \text{and} \quad T(2) = 9 \quad \text{but} \quad T(3) = 11 \neq T(1) + T(2) ,$$

so T is not linear.

Warning! In school you probably called this kind of function linear – it is **not** linear in the sense of linear algebra.

Example 8. Let $T : \mathbb{R} \rightarrow \mathbb{R}$ where $T(x) = \sin x$. Is this function a linear map?

$$T(\pi) = 0 \quad \text{but} \quad T\left(\frac{1}{2}\pi\right) = 1 \neq \frac{1}{2} T(\pi) .$$

So T is not a linear transformation.

Example 9. Let $T : \mathbb{R} \rightarrow \mathbb{R}$ where $T(x) = \sqrt{x}$. Is this function a linear map?

$$T(1+1) = T(2) = \sqrt{2} \quad \text{but} \quad T(1) + T(1) = 1 + 1 = 2 \neq T(1+1) .$$

So T is not linear.

Comment. Apart from the algebra, there is something much more seriously wrong with this example. What is it?

ALL linear maps do that.

Fact (Linear maps preserve zero and negatives)

Let $T : V \rightarrow W$ be linear. Then

1. $T(\vec{0}) = \vec{0}$;
2. for any $\vec{v} \in V$ we have $T(-\vec{v}) = -T(\vec{v})$.

Proof. Use some results we proved in Chapter 6, together with the fact that T preserves scalar multiplication: we have

$$T(\vec{0}) = T(0\vec{v}) = 0 T(\vec{v}) = \vec{0}$$

$$\text{and } T(-\vec{v}) = T((-1)\vec{v}) = (-1)T(\vec{v}) = -T(\vec{v}).$$

Exercise. Give alternative proofs of these results, using properties of vector addition instead of scalar multiplication. *Hint*: consider $T(\vec{0} + \vec{0})$ and $T(\vec{v} + (-\vec{v}))$.

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What is this good for?



These two properties can sometimes be used to show that a function is *not* linear: ALL linear maps preserve zero and negatives, so a function which does *not* preserve zero or does *not* preserve negatives *cannot* be linear.

In practice: If you are wondering if a function is linear, first test whether it satisfies these two conditions. If not, it is not linear, end of the story.

Using these properties to show that a function is NOT linear. Examples.

- a) In example 7 we showed that $T : \mathbb{R} \rightarrow \mathbb{R}$ given by $T(x) = 2x + 5$ is not linear because it does not preserve addition. Alternatively, since $T(\mathbf{0}) = 5 \neq 0$, we can instantly say that T is not linear.

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b) Define $T : \mathbb{R} \rightarrow \mathbb{R}^2$ by $T(x) = (x^2, x^3)$. Then

$$T(2) = (4, 8) \quad \text{but} \quad T(-2) = (4, -8) \neq -T(2),$$

so T is not linear.

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c) As in example 8, consider $T : \mathbb{R} \rightarrow \mathbb{R}$ given by $T(x) = \sin x$. Then

$$T(0) = 0 \quad \text{and} \quad T(-x) = -T(x) \text{ for every } x.$$

What does this tell us?

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What does this tell us? Nothing. It does not tell us that the function is linear and it does not tell us either that the function is not linear. Bummer.

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Comments.

- These properties can never be used to show that a function is linear. As shown in the last example, $T(x) = \sin x$ passes both tests, yet we already know that it is not linear.

In other words, preserving zero and negatives is a necessary condition for a function to be linear, but it is not a sufficient condition.

- In using the “negative” test, always give a specific numerical example.



A badly written solution.

Example 10. Criticise the following linear transformation proof. Find all the mistakes, as well as pointing out any parts which could have been expressed better.

Problem. Show that $T : \mathbb{R}^4 \rightarrow \mathbb{R}$ given by $T(\vec{x}) = x_1 - 6x_2 + x_4$ is a linear mapping.

'Solution'. Let $\vec{x}, \vec{y} \in T$. Then

$$\begin{aligned}T(\vec{x} + \vec{y}) &= (x_1 + y_1) - 6(x_2 + y_2) + (x_4 + y_4) \\&= (x_1 - 6x_2 + x_4) + (y_1 - 6y_2 + y_4) \\&= T(\vec{x}) + T(\vec{y}),\end{aligned}$$

so T is closed under addition.

Let $\lambda \in \mathbb{R}$. Then

$$\begin{aligned}T(\lambda \vec{x}) &= \lambda T(\vec{x}) \quad \Rightarrow \quad T(\lambda x_1, \lambda x_2, \lambda x_4) = \lambda T(x_1, x_2, x_4) \\&\Rightarrow (\lambda x_1) - 6(\lambda x_2) + (\lambda x_4) = \lambda(x_1 - 6x_2 + x_4) \\&\Rightarrow 0 = 0,\end{aligned}$$

so T is closed under multiplication.

Since T is closed under addition and multiplication, it is linear.

Linear maps preserve linear combinations

The conditions on vector addition and scalar multiplication in the definition of a linear transformation can be combined into a single statement.

Theorem (Linear maps preserve linear combinations)

A function $T : V \rightarrow W$ between vector spaces is linear if and only if

$$T(\lambda_1 \vec{v}_1 + \lambda_2 \vec{v}_2) = \lambda_1 T(\vec{v}_1) + \lambda_2 T(\vec{v}_2) \quad (*)$$

for all vectors $\vec{v}_1, \vec{v}_2$ in V and all scalars λ_1, λ_2 .

Proof. Suppose that T is linear. Since T preserves addition and scalar multiplication we have

$$T(\lambda_1 \vec{v}_1 + \lambda_2 \vec{v}_2) = T(\lambda_1 \vec{v}_1) + T(\lambda_2 \vec{v}_2) = \lambda_1 T(\vec{v}_1) + \lambda_2 T(\vec{v}_2)$$

for all vectors $\vec{v}_1, \vec{v}_2$ and scalars λ_1, λ_2 . Conversely, suppose that T satisfies $(*)$ for all vectors $\vec{v}_1, \vec{v}_2$ and scalars λ_1, λ_2 . By taking $\lambda_1 = \lambda_2 = 1$ we see that T preserves addition; by taking $\lambda_2 = 0$ we see that T preserves scalar multiplication. Therefore, T is linear.



But what if there are more terms in the linear combination?

Linear maps preserve linear combinations

Theorem (**Linear maps preserve longer linear combinations**)

If $T : V \rightarrow W$ is linear then

$$T(\lambda_1 \vec{v}_1 + \lambda_2 \vec{v}_2 + \cdots + \lambda_n \vec{v}_n) = \lambda_1 T(\vec{v}_1) + \lambda_2 T(\vec{v}_2) + \cdots + \lambda_n T(\vec{v}_n)$$

for all $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$ in V and all scalars $\lambda_1, \lambda_2, \dots, \lambda_n$.

Proof.

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Proof. Proof by induction using previous result.

Linear maps and images of a basis

Example 11. Let T be a linear map from $\mathbb{R}^3$ to $\mathbb{R}^2$, with

$$T(1, 0, 0) = (1, 2), \quad T(0, 1, 0) = (3, 4), \quad T(0, 0, 1) = (5, 6).$$

a) **Warm-up.** Find $T(3, 1, -2)$.

b) **Now the general case.** For any $\vec{x} = (x_1, x_2, x_3)^T$ in $\mathbb{R}^3$, find the expression of $T(\vec{x})$.

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a) Find $T(3, 1, -2)$.



b) For any $\vec{x}$ in $\mathbb{R}^3$ we have

$$\begin{aligned} T(\vec{x}) &= T(x_1, x_2, x_3) = T(x_1(1, 0, 0) + x_2(0, 1, 0) + x_3(0, 0, 1)) \\ &= x_1 T(1, 0, 0) + x_2 T(0, 1, 0) + x_3 T(0, 0, 1) \\ &= x_1(1, 2) + x_2(3, 4) + x_3(5, 6) \\ &= (x_1 + 3x_2 + 5x_3, 2x_1 + 4x_2 + 6x_3). \end{aligned}$$

Values on a basis is all we need if the function is linear.

Observe that example 11 taught us something very important!

Theorem (A linear transformation is entirely determined by its values on a basis)

Suppose that $T : V \rightarrow W$ is linear, let $B = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ be a basis for V , and let $\mathbf{v}$ be a vector in V .

If $x_1, x_2, \dots, x_n$ are scalars such that $\mathbf{v} = x_1\mathbf{b}_1 + x_2\mathbf{b}_2 + \dots + x_n\mathbf{b}_n$, then

$$T(\mathbf{v}) = x_1 T(\mathbf{b}_1) + x_2 T(\mathbf{b}_2) + \dots + x_n T(\mathbf{b}_n) . \quad (*)$$

Proof. This follows immediately from the the fact that linear maps preserve linear combinations.

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Proof. This follows immediately from the the fact that linear maps preserve linear combinations.



The point of this result is that if T is a linear map with domain V and $\{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ is a basis for V , and if we know the values of $T(\mathbf{b}_1), T(\mathbf{b}_2), \dots, T(\mathbf{b}_n)$, then we can calculate $T(\mathbf{v})$ for *all* vectors $\mathbf{v}$ in V . Given any vectore $\mathbf{v}$, we can solve linear equations to find scalars $x_1, x_2, \dots, x_n$ such that $\mathbf{v} = x_1\mathbf{b}_1 + x_2\mathbf{b}_2 + \dots + x_n\mathbf{b}_n$; and then $(*)$ gives the value of $T(\mathbf{v})$.

Values on a basis is all we need if the function is linear.

Example 12. Let $T : \mathbb{P}_3 \rightarrow \mathbb{R}$ be linear; suppose it is known that

$$T(\mathbf{1}) = 5, \quad T(\mathbf{x}) = 2, \quad T(\mathbf{x}^2) = 3 \quad \text{and} \quad T(\mathbf{x}^3) = -1.$$

Find an expression for $T(p)$ valid for any p in $\mathbb{P}_3$.

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 We can compute $T(p)$ for any p in $\mathbb{P}_3$, because

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 We can compute $T(p)$ for any p in $\mathbb{P}_3$, because

$$\begin{aligned} T(a + bx + cx^2 + dx^3) &= T(a\mathbf{1} + b\mathbf{x} + c\mathbf{x}^2 + d\mathbf{x}^3) \\ &= aT(\mathbf{1}) + bT(\mathbf{x}) + cT(\mathbf{x}^2) + dT(\mathbf{x}^3) \\ &= 5a + 2b + 3c - d. \end{aligned}$$